

SYLLABUS

COGNITION AND DECISION MAKING

01:185:301 Section: E1 (00708)
Summer 2025

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COURSE OVERVIEW

Logistics

Main page: <https://canvas.rutgers.edu>

Pre-requisites: Cognitive Science: A Multi-disciplinary Introduction (01:185:201)

Lectures:

- Asynchronous (pre-recorded) via Zoom

Recitations:

- Once per week via Zoom (dates and times to be determined after class feedback)

Instructor: Paul Robinson

- Email: pdr49@ruccs.rutgers.edu (response within 1 working day)
- Office hours / student support hours: By appointment via Zoom

Course description

Which subject should I major in? Should I eat meat? Who should I vote for? Should I get vaccinated? Which stocks should I buy? Should I overlook unethical behavior by my employer?

The length and quality of your life is largely determined by the decisions you and others make. Human choices also impact animals and the environment. What is it for a decision to be rational or irrational? Why do we make the decisions we do? Can knowledge of how our minds work be used to manipulate our decisions? These are some of most pressing questions of our time.

This course addresses these kinds of questions about decision-making from the interdisciplinary perspective of cognitive science. It uses knowledge and techniques from disciplines such as psychology, neuroscience, anthropology, economics, and philosophy, to investigate the “mental machinery” that underlies human decision-making. Topics covered include unconscious influences on decisions, heuristics that people use to make decisions, and systematic biases displayed by decisions. The course concludes by examining how we can improve decision-making. Examples are drawn from domains such as marketing, finance, healthcare, public policy, and everyday life. Throughout the course there will be a focus on (i) understanding the mathematical formalisms that guide research on decision-making, such as Bayesian probability theory and expected utility theory, and (ii) critically evaluating the research methods and experimental paradigms employed by classic studies of decision-making.

Core Curriculum Learning Goals



This course fulfills the “**Quantitative and Formal Reasoning**” goal (QQ): You will formulate, evaluate, and communicate conclusions and inferences from quantitative information.

Department Learning Goals

Upon successful completion of the course, students will:

- i. Appreciate the interdisciplinary nature of cognitive science and be able to read, summarize, and critically evaluate scientific research papers from multiple disciplines.
- ii. Understand the factors that influence decision making, the biases that decisions tend to display, and theories of the mental processes that underlie decision-making.
- iii. Develop strategies for making better decisions.

HOW THIS COURSE WORKS

Mode of delivery

This course is asynchronous (pre-recorded) via Zoom.

The course follows a weekly schedule but each week you can view the lecture recordings at your own pace.

There will be online quizzes on lecture content.

Attendance policy

Although there will be weekly synchronous recitation meetings, attendance is not required.

Course requirements

- **Reading and lecture quizzes:** Each week, by Friday at 11:59pm, you will use Canvas to complete brief multiple-choice quizzes on the readings and the lectures for that week. Each quiz will open on Monday at 9am, giving you five days to do the reading and watch the lectures.
- **Module reflection posts:** This course is divided into five weekly modules. By 11:59pm on Saturday you will submit a reflection post to a discussion board on Canvas. These posts will be used for recitation discussion and will also help you choose a topic for your final paper. Your reflection should consist of four sentences: (1) state the most interesting thing that you learned; (2) explain why you find it interesting; (3) ask a question concerning something you didn't understand or are unsure about; (4) explain why you didn't understand or why you are unsure.
- **Writing assignments:** You will submit four short writing assignments. Further details will be provided in Canvas. The first three are reflection pieces and the final one is designed to help you with your final paper:

Assignment A: Analyze one good decision and one bad decision that you made in the past year.

Assignment B: Write a memo to your future boss explaining how the cognitive science of decision-making relates to your imagined future profession.

Assignment C: Design an educational poster on the communication of uncertainties.

Assignment D: Develop a brief literature review – of the type that might be found in the introductory section of a journal article – on a course-related subject of your choice. (This assignment may serve as the foundation for your final paper).

- **Mathematical homework assignments:** There will be five mathematics assignments on probability theory, expected utility theory, and game theory.
- **Midterm exam:** Using Canvas, you will take a multiple-choice and short-answer quiz.
- **Final examination:** Using Canvas, you will take a multiple-choice and short-answer quiz. You will also submit a short research paper on a course-related topic of your choosing.

How your final grade is calculated

ASSIGNMENT CATEGORY	PERCENTAGE
Reflection posts	10%
Reading quizzes	10%
Lecture quizzes	10%
Math homeworks	10%
Writing assignments	10%
Midterm exam quiz	15%
Final exam quiz	15%
Final paper	20%
Total	100

I use the following grade scale, where $[x - y)$ denotes values from x up to but not including y :

$[90 - 100]$: A
 $[85 - 90)$: B+
 $[80 - 85)$: B
 $[75 - 80)$: C+
 $[70 - 75)$: C
 $[60 - 70)$: D
 Below 60: F

Late assignments

If you experience a situation that prevents you from completing work on time, please inform me as soon as possible. Late work will be given a score of “0” on Canvas by default. However, I am happy to consider both granting extensions and giving you full or partial credit for late work (by manually overriding the default 0 grade on Canvas), depending on the circumstances. Please do not hesitate to contact me about any aspect of the course that you are having difficulty with. I am here to help you learn and succeed in the course, not to make your life more difficult.

There will be no personalized end-of-semester extra-credit assignments. If you have concerns about your final grade, please speak to me early in the semester.

Required reading

There is one required textbook for this class:

- Kim, Nancy. (2017). *Judgment and Decision-Making: In the Lab and the World*. Macmillan International Higher Education. ISBN: 9781137269560, 1137269561

Copies are available in the Rutgers bookstore.

Other required readings will be uploaded to Canvas in pdf form.

RIGHTS AND RESPONSIBILITIES

Academic integrity policy

Rutgers University takes academic dishonesty very seriously. By enrolling in this course, you assume responsibility for familiarizing yourself with the Academic Integrity Policy and the possible penalties (including suspension and expulsion) for violating the policy. As per the policy, all suspected violations will be reported to the Office of Student Conduct.

Academic dishonesty includes (but is not limited to):

- Cheating
- Plagiarism
- Aiding others in committing a violation or allowing others to use your work
- Failure to cite sources correctly
- Fabrication
- Using another person's ideas or words without attribution—re-using a previous assignment
- Unauthorized collaboration
- Sabotaging another student's work

If in doubt, please consult the instructor. Please review the Academic Integrity Policy:

<http://nbacademicintegrity.rutgers.edu/home/academic-integrity-policy/>

Copyright disclaimer

Almost all original work is the intellectual property of its authors. This includes not just books and articles, but the syllabi, lectures, lecture slides, recorded lectures, course materials, presentations, homework problems, exams, and other materials used in this course, in either printed or electronic form.

Providing course materials to commercial suppliers such as CourseHero, Chegg, etc. and/or publicly distributing or displaying course materials, or helping others to do so, is a violation of academic integrity.

The authors hold copyrights in their works, which are protected by U.S. statutes. Copying this work or posting it online without the permission of the author may violate the author's rights. More importantly, these works are the product of the author's efforts; respect for these efforts and for the author's intellectual property rights is an important value that members of the university community take seriously. For more instructions on copyright protections at Rutgers University, please refer to:

<https://www.libraries.rutgers.edu/research-tools-and-services/copyright-guidance/copyright-students>

Requesting accommodations

Rutgers students are expected to attend all scheduled course meetings. University policy excuses absences due to religious observance or participation in Rutgers-approved activities, and permits students to make up work missed for these circumstances. If you will be absent from a class, lab, or exam for any reason, please report your absence here:

<https://sims.rutgers.edu/ssra/>

If you have been told to quarantine, or are experiencing symptoms of any transmittable disease, please remain at home and do not attend in-person class meetings. If you need help to catch up on material you missed, I am happy to schedule personal sessions with you.

Rutgers University welcomes students with disabilities into all the University's educational programs. In order to receive consideration for reasonable accommodations, a student with a disability must contact the appropriate disability services office at the campus where you are officially enrolled, participate in an intake interview, and provide documentation:

<https://ods.rutgers.edu/students/documentation-guidelines>.

If the documentation supports your request for reasonable accommodations, your campus's disability services office will provide you with a Letter of Accommodations. Please share this letter with your instructors and discuss the accommodations with them as early in your courses as possible. To begin this process, please complete the Registration form on the ODS web site at: <https://ods.rutgers.edu/students/registration-form>.

Full disability policies and procedures are at: (848) 445-6800 / Lucy Stone Hall, Suite A145, Livingston Campus, 54 Joyce Kilmer Avenue, Piscataway, NJ 08854 / <https://ods.rutgers.edu/>

Accessibility of course technology

This course requires use of Canvas and other online communication and multimedia tools. If you need additional services to use these technologies, please request accommodations with your instructor.

Please visit the [Rutgers Student Tech Guide](#) page for resources available to all students. If you do not have the appropriate technology for financial reasons, please email Dean of Students deanofstudents@echo.rutgers.edu for assistance.

If you are facing other financial hardships, please visit the Office of Financial Aid at <https://financialaid.rutgers.edu/>.

Student wellness resources

- Student Success Essentials: <https://success.rutgers.edu>
- Student Support Services: <https://www.rutgers.edu/academics/student-support>
- The Learning Centers: <https://rlc.rutgers.edu/>
- Rutgers Libraries: <https://www.libraries.rutgers.edu/>
- Bias Incident Reporting: <https://studentaffairs.rutgers.edu/bias-incident-reporting>
- Dean of Students – Student Support Office: <https://success.rutgers.edu/resource/dean-students-student-support-office>
- Office of Veteran and Military Programs and Services: <https://veterans.rutgers.edu>
- Student Health Services: <http://health.rutgers.edu/>
- Counseling, Alcohol and Other Drug Assistance Program & Psychiatric Services (CAPS): <http://health.rutgers.edu/medical-counseling-services/counseling/>
- UWill: free immediate access to teletherapy; you can choose a therapist based on your preferences including issue, gender, language, ethnicity. <http://health.rutgers.edu/uwill/>
- Office for Violence Prevention and Victim Assistance: www.vpva.rutgers.edu/
- Office of Disability Services: <https://ods.rutgers.edu/>
- Basic Needs Assistance (food, housing, and other essentials): <https://ruoffcampus.rutgers.edu/basic-needs>
- Rutgers Student Food Pantry: <https://ruoffcampus.rutgers.edu/food-pantry>

COURSE SCHEDULE

Module A: Heuristics and Biases

Week 1 material

Reading	Kim (2018): Introduction; Availability and representativeness; Anchoring and primacy effects in judgment.
Lectures	Dual-process theories. Availability and representativeness. Anchoring and primacy effects. Intro to probability theory.

Week 1 schedule

Tuesday, May 27	
Wednesday, May 28	
Thursday, May 29	
Friday, May 30	Lecture and reading quizzes due
Saturday, May 31	Module A reflection post due and math homework 1 due
Sunday, June 1	Writing assignment A due

Module B: Learning and Predicting

Week 2 material

Reading	Kim (2018): Hindsight bias; Risk perception; Prediction; Choice and mental accounting.
Lectures	Learning from past decisions. Risk analysis. Choice architecture. Bayes theorem.

Week 2 schedule

Monday, June 2	
Tuesday, June 3	
Wednesday, June 4	
Thursday, June 5	
Friday, June 6	Lecture and reading quizzes due
Saturday, June 7	Module B reflection post due and math homework 2 due
Sunday, June 8	Writing assignment B due

Module C: Framing Information

Week 3 material

Reading	Kim (2018): Expected utility theory; Framing effects and prospect theory.
Lectures	Framing effects. Expected utility theory.

Week 3 schedule

Monday, June 9	
Tuesday, June 10	
Wednesday, June 11	
Thursday, June 12	
Friday, June 13	Lecture and reading quizzes due
Saturday, June 14	Module C reflection post due and math homework 3 due
Sunday, June 15	Midterm exam due

Module D: Modeling Reality

Week 4 material

Reading	Kim (2018): Schemas & framework theories; Judging covariation, contingency & cause; Hypothesis testing and confirmation bias; Belief.
	Explanation-based decisions.
Lectures	Rationalizing decisions. Game theory.

Week 4 schedule

Monday, June 16	
Tuesday, June 17	
Wednesday, June 18	
Thursday, June 19	
Friday, June 20	Lecture and reading quizzes due
Saturday, June 21	Module D reflection post due and math homework 4 due
Sunday, June 22	Writing assignment C due

Module E: Society and Groups

Week 5 material

Reading	Newell, Lagnado, Shanks (2015): Group decision making. Kim (2018): Moral judgment and cooperation.
Lectures	Group decision making. Moral decision making.

Week 5 schedule

Monday, June 23	
Tuesday, June 24	
Wednesday, June 25	
Thursday, June 26	
Friday, June 27	Lecture and reading quizzes due
Saturday, June 28	Module E reflection post due and optional math homework 5 due
Sunday, June 29	Writing assignment D due

Final Exam

Week 6 material

Reading	None
Lectures	None

Week 6 schedule

Monday, June 30	Final exam multiple choice quiz due
Tuesday, July 1	
Wednesday, July 2	Final exam paper due
Thursday, July 3	
Friday, July 4	